Fall 2024 Sql Databases

Module 8 – Exercises

Exercise (1)

- In the global market, businesses face diverse customer behaviors that can vary significantly across countries. One of the key factors in understanding these behaviors is the variance in payment amounts made by customers. Variance measures the extent to which individual payments deviate from the average payment within a country. By examining this variance, businesses can gain insights into the spending patterns of their customers, which can help tailor pricing strategies, optimize payment terms, and improve promotional campaigns for different regions.
- This report will answer the following questions:
- **What is the variance in payment amounts made by customers from different countries?**
- **How does the variance in payment amounts differ across countries? What insights can be drawn from this difference in variance?**
- **What is the significance of variance in understanding customer payment behaviors?**

- **<u>Note: using classicalmodel schema</u>**

How does the variance in payment amounts differ across countries? What insights can be drawn from this difference in variance?

- The variance of payment amounts across different countries reveals the level of payment predictability in each region. A high variance indicates that customer payments are more volatile and less predictable, while a low variance suggests more consistency in payment amounts.

- **Analysis of the Variance Differences:**
- **High Variance**

- In countries with high variance, customers may make large payments at irregular intervals or show a wide range of spending behaviors. This could be indicative of certain customers making high-value purchases or others making smaller, more frequent payments.

- **Low Variance**
- Countries with low variance show more stable and consistent spending behaviors. Customers in these regions are more predictable in their payment patterns, making it easier for businesses to forecast revenue and plan accordingly.

Solution (1): Analysis of the Variance Differences:

- What insights can be drawn from this difference in variance?

SELECT c.country, **VARIANCE(p.amount)** AS PaymentVariance
FROM classicmodels.payments pJOIN classicmodels.customers c ON p.customerNumber = c.customerNumberGROUP BY c.country;

Sample of the result :

country	PaymentVariance
France	289091255.61852616
USA	516230148.1543597
Australia	307095509.32688594
Norway	331957250.0432187
Germany	270101856.7593102
Spain	812331482.7927732
Sweden	223092980.34306878

Analysis of the Variance Differences:

•**Identifying Payment Patterns**: Variance helps businesses identify trends and anomalies in payment behaviors.
•A high variance suggests that customers exhibit diverse payment habits, which might include different amounts, payment methods, or frequencies.
•A low variance indicates that customers follow more similar payment patterns.

•Variance in customer payment behaviors helps businesses identify patterns and anomalies, forecast payment trends, manage cash flow, segment customers, and tailor payment strategies. High variance indicates diverse payment habits, while low variance suggests consistency. Understanding variance allows businesses to optimize payment systems, mitigate risks, and improve financial stability by adjusting credit terms, payment options, and customer segmentation

Exercise (2)

- RetailMax, a retail company, wants to analyze its sales and payment data for the past quarter. They aim to evaluate their product performance, identify high and low-performing items, and assess the consistency in sales to make more informed inventory and marketing decisions.

- **The company has two main tables to analyze:**

- **Payments Table:**

Contains details of payments made by customers.

Columns: customerNumber, checkNumber, paymentDate, amount

- **OrderDetails Table:**

Contains details of the products ordered by customers.

Columns: orderNumber, productCode, quantityOrdered,

priceEach, orderLineNumber

Task 1:
Mean Payment Amount:

Using the sales data from the payments and orderdetails tables, answer the following:

1- What is the average payment amount made by customers in the last quarter?
This will help RetailMax understand the general trend of customer spending.

```
SELECT AVG(amount) AS mean_paymentFROM sakila.payment
WHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01’;
```

mean_payment
4.217858

Task 2
Max Payment Amount:

- What is the highest payment amount made by a customer?
- This will highlight high-value transactions and help identify the top-spending customers.
- SELECT MAX(amount) AS max_payment
- FROM sakila.payment
- WHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01’;

	max_payment
▶	11.99

Task 3:
Min Payment Amount:

- What is the lowest payment amount?
- Understanding the lowest transaction can help RetailMax identify low-spending customers or small-value transactions.

```
SELECT MIN(amount) AS min_payment
FROM sakila.payment
WHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01';
```

	min_payment
▶	0.99

Task 4
Range of Payment Amounts:

- What is the difference between the highest and lowest payment amounts made by customers?
- The range will show how much variation exists in customer payments and can assist in determining if there is a significant difference between large and small transactions.

```
SELECT (MAX(amount) - MIN(amount)) AS payment_rangeFROM
sakila.payment
WHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01’;
```

	payment_range
▶	11.00

Task 5
Standard Deviation of Payment Amounts:

- What is the standard deviation of the payment amounts?
- This metric will show how much variability exists in the payment amounts, which will help RetailMax understand how consistent or inconsistent customer payments are.

```
SELECT STDDEV(amount) AS payment_stddevFROM sakila.payment
WHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01';
```

	payment_stddev
▶	2.3557626776541247

Task 6
Variance of Payment Amounts:

- What is the variance of the payment amounts?
- This will indicate how widely payments differ from the average payment and highlight any large fluctuations in customer spending.

payment_variance
5.549617793428131

- SELECT VARIANCE(amount) AS payment_varianceFROM sakila.paymentWHERE payment_date BETWEEN '2005-07-09' AND '2005-08-01’;

payment_variance
5.549617793428131

Exercise (3)

- **RetailMax Entertainment**, a video rental and sales company, wants to optimize its product offerings based on the performance of its films. The company needs a detailed analysis of how its films are performing in terms of revenue during a specific period.

- The company has extensive sales and rental data, but they lack an overview of how each film is contributing to their revenue. They want to know the total sales revenue for each film during a specified period to help guide their decisions.

Calculate Total Revenue by Film:

- SELECT f.title,
-     SUM(p.amount) AS total_sales_revenue
- FROM sakila.rental r
- JOIN sakila.payment p ON r.rental_id = p.rental_id -- Join rental with payment on rental_id
- JOIN sakila.inventory i ON r.inventory_id = i.inventory_id -- Join rental with inventory to get film info
- JOIN sakila.film f ON i.film_id = f.film_id -- Join inventory with film to get the title
- WHERE p.payment_date BETWEEN '2005-07-09' AND '2005-08-01' -- Date range filter
- GROUP BY f.title -- Group by film title to calculate revenue for each film
- ORDER BY total_sales_revenue DESC; -- Sort by highest revenue

title	total_sales_revenue
SATURDAY LAMBS	77.91
INNOCENT USUAL	77.90
DOGMA FAMILY	77.87
HUSTLER PARTY	76.92
ZORRO ARK	76.90
TITANS JERK	73.89
SCORPION APOLLO	69.91
TORQUE BOUND	68.91
CONFIDENTIAL INTERVIEW	68.88
METROPOLIS COMA	67.88
CLOSER BANG	65.90
SCALAWAG DUCK	65.88
MOONSHINE CABIN	64.90
SATISFACTION CONFIDEN...	63.91
FLAMINGOS CONNECTICUT	63.90
LADY STAGE	62.91
NIGHTMARE CHILL	62.91
HARRY IDAHO	62.90
FOOL MOCKINGBIRD	60.92
WIFE TURN	60.91
GOODFELLAS SALUTE	60.91
SUIT WALLS	60.91
RANGE MOONWALKER	60.90
TIMBERLAND SKY	60.88
ROCKY WAR	59.92
WHISPERER GIANT	59.91
LIES TREATMENT	59.89
TITANIC BOONDOCK	58.91
STAGECOACH ARMAGEDDON	58.90
DRIFTER COMMANDMENTS	58.90
SABRINA MIDNIGHT	58.89
FLINTSTONES HAPPINESS	57.93
BIRDS PERDITION	57.91
TRIP NEWTON	57.89

By identifying which films generate the most revenue, the company can quickly identify **high-performing films**. This helps them focus marketing efforts, promotions, and inventory decisions on popular films.

By examining the total revenue per film, the company can identify which films **attract the most paying customers** and which films have the highest customer engagement.

Exercise (4)
Analyzing Sales Over Time

- Farmers Market wants to better understand the seasonal trends in customer purchases. They are particularly interested in identifying any fluctuations in sales volume over time and the impact of seasonal factors on their sales revenue. By analyzing sales data from the past year, they hope to identify which months generate the highest revenue and which months are slower, enabling them to better plan for inventory, promotions, and staffing needs.

- Using the following query, analyze the monthly sales trends over the past year. Calculate the total sales for each month, and identify the peak sales months, as well as any months with significantly lower sales. Provide insights that can help Farmers Market optimize their operations.

Solution (4)

```
SELECT
  EXTRACT(YEAR FROM market_date) AS sales_year,
  EXTRACT(MONTH FROM market_date) AS sales_month,
  SUM(cost_to_customer_per_qty*quantity) AS total_monthly_sales
FROM
  farmers_market.customer_purchases
GROUP BY
  EXTRACT(YEAR FROM market_date),
  EXTRACT(MONTH FROM market_date)
ORDER BY
  sales_year ASC,
  sales_month ASC;
```

Solution (4)

sales_year	sales_month	total_monthly_sales
2019	4	1900.00
2019	5	2150.00
2019	6	1985.05
2019	7	1826.65
2019	8	1967.45
2019	9	1731.59
2019	10	2096.50
2019	11	2704.50
2019	12	2415.50
2020	3	1809.00
2020	4	1544.50
2020	5	2047.00
2020	6	1758.95
2020	7	1703.12
2020	8	1981.63
2020	9	1946.15
2020	10	752.00

Exercise (5)

The farmers' market vendor management team wants to improve its sales forecasting and inventory planning. Currently, the sales data is recorded in the customer_purchases table, which tracks each sale's product, quantity, and cost to the customer.

The team has identified the need to compare the total sales quantity and total sales value for each product sold by vendors, broken down by market day. This will help them understand which products are performing well, assess vendor performance, and ensure that the correct inventory levels are maintained for future market days.

Solution (5)

```
SELECT cp.product_id, cp.vendor_id, cp.market_date, COALESCE(SUM(cp.quantity), 0) AS total_sales_quantity, COALESCE(SUM(cp.quantity * cp.cost_to_customer_per_qty), 0) AS total_sales_value
FROM farmers_market.customer_purchases cp
GROUP BY cp.product_id, cp.vendor_id, cp.market_dateORDER BY cp.product_id, cp.vendor_id, cp.market_date;
```

Solution (5)

product_id	vendor_id	market_date	total_sales_quantity	total_sales_value
1	7	2019-07-03	7.38	51.5862
1	7	2019-07-06	10.96	76.6104
1	7	2019-07-10	10.93	76.4007
1	7	2019-07-13	10.22	71.4378
1	7	2019-07-17	7.14	49.9086
1	7	2019-07-20	9.04	63.1896
1	7	2019-07-24	3.13	21.8787
1	7	2019-07-27	5.74	40.1226
1	7	2019-07-31	11.23	78.4977
1	7	2019-08-03	2.49	17.4051
1	7	2019-08-07	4.96	34.6704
1	7	2019-08-10	10.73	75.0027
1	7	2019-08-14	10.63	74.3037
1	7	2019-08-21	1.59	11.1141
1	7	2019-08-24	7.34	51.3066
1	7	2019-08-28	4.75	33.2025
1	7	2019-08-31	7.40	51.7260
1	7	2019-09-04	10.13	70.8087
1	7	2019-09-07	7.09	49.5591
1	7	2019-09-11	8.84	61.7916
1	7	2019-09-14	7.08	49.4892
1	7	2019-09-18	2.38	16.6362
1	7	2019-09-21	8.21	57.3879
1	7	2019-09-25	2.22	15.5178
1	7	2019-09-28	3.57	24.9543
1	7	2020-07-01	10.29	71.9271
1	7	2020-07-04	9.13	63.8187
1	7	2020-07-08	11.85	82.8315
1	7	2020-07-11	10.08	70.4592
1	7	2020-07-15	10.01	69.9699
1	7	2020-07-18	4.66	32.5734
1	7	2020-07-22	8.52	59.5548
1	7	2020-07-25	10.29	71.9271
1	7	2020-08-01	0.50	3.4950

Exercise (6)

- The vendor management team at the farmers' market wants to identify product categories that have not made any sales during a specific period. This information is crucial for assessing product performance and understanding which categories may need more marketing efforts or product promotions.
- Write an SQL query that returns the list of product categories that have no sales at all, including the count of sales for each category. Ensure that categories with no sales are clearly identified by a sales count of 0, and order the results by the product category ID.

Solution (6)

- -- Query to identify categories with no sales at all

SELECT p.product_category_id, -- Category of the product (assuming this is the correct column) COUNT(cp.product_id) AS sales_count -- Count of sales for each category

FROM

farmers_market.product pLEFT JOIN farmers_market.customer_purchases cp ON p.product_id = cp.product_idGROUP BY p.product_category_id -- Group by product categoryHAVING COUNT(cp.product_id) = 0 -- Categories with no sales (NULL values for sales)ORDER BY p.product_category_id; -- Order by product category ID

product_category_id	sales_count
2	0
5	0
6	0
7	0